Lesson 1: Measure Spaces

σ-Algebras, Measures, and Measurable Functions

Session 1 of 102 | Phase I: Mathematical Foundations | 8 h
Primary text: Durrett, *Probability: Theory and Examples*, Chapter 1

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

February 9, 2026

Abstract

This lesson develops the measure-theoretic foundations needed for rigorous probability and, ultimately, for the convergence proofs of reinforcement learning algorithms. We construct the framework of σ-algebras and measures from first principles, prove Carathéodory's extension theorem, define measurable functions and their approximation by simple functions, and connect everything to probability theory. Every result is proved. The reader who masters this material will have the language and tools to define conditional expectation, filtrations, and martingales rigorously — the machinery that drives all stochastic approximation proofs in RL.

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1 Why Measure Theory?

The convergence proofs of TD-learning, Q-learning, and policy gradient methods rest on stochastic approximation theory, which in turn rests on martingale convergence theorems. To state these theorems precisely, we need:

1. **σ-algebras** to model the information available at each time step (filtrations).
2. **Measures** to assign probabilities to events in a mathematically consistent way.
3. **Measurable functions** to formalize random variables and value functions.
4. **Integration** (next lesson) to define expectations rigorously.

Without measure theory, "$\mathbb{E}[X \mid \mathcal{F}_t]$" is just notation. With it, conditional expectation becomes a well-defined random variable characterized by two precise properties, and the entire convergence machinery follows.

Why This Matters for RL

The TD(0) convergence proof (Theorem 4.2 in the TD article) uses:

- The filtration $\mathcal{F}_t = \sigma(S_0, A_0, R_1, \ldots, S_t)$ — a σ-algebra generated by random variables.
- The noise term $w_k(s)$ satisfying $\mathbb{E}[w_k(s) \mid \mathcal{F}_k] = 0$ — conditional expectation with respect to a σ-algebra.
- The Robbins–Siegmund theorem, which requires a *non-negative supermartingale* — defined via σ-algebras and conditional expectation.

This lesson provides the language for all three.

2 σ-Algebras

2.1 Definition and First Properties

Throughout, Ω denotes a non-empty set (the *sample space*).

Definition 2.1 (σ-Algebra)**.** A collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ of subsets of Ω is a *σ-algebra* (or *σ-field*) on Ω if:

(i) $\Omega \in \mathcal{F}$.

(ii) If $A \in \mathcal{F}$, then $A^c := \Omega \setminus A \in \mathcal{F}$. (closure under complement)

(iii) If $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. (closure under countable union)

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. Elements of $\mathcal{F}$ are called *measurable sets* (or *events* in probability).

Remark 2.1 (Why Countable, Not Arbitrary?). Axiom (iii) requires closure under *countable* unions, not arbitrary ones. Closure under arbitrary unions would force $\mathcal{F} = \mathcal{P}(\Omega)$ in most interesting cases, which is too large: as we will see (Theorem 4.4), the power set of $\mathbb{R}$ admits no translation-invariant measure extending the notion of "length." The restriction to countable operations is precisely what makes non-trivial measures possible.

Proposition 2.1 (Basic Properties)**.** *Let $\mathcal{F}$ be a σ-algebra on Ω. Then:*

(a) $\emptyset \in \mathcal{F}$.

(b) If $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$. (closure under countable intersection)

(c) If $A, B \in \mathcal{F}$, then $A \setminus B \in \mathcal{F}$ and $A \triangle B \in \mathcal{F}$. (closure under difference and symmetric difference)

(d) $\mathcal{F}$ is closed under finite unions and finite intersections.

Proof. **(a)** $\emptyset = \Omega^c \in \mathcal{F}$ by (i) and (ii).

(b) By De Morgan's law: $\bigcap_{n=1}^{\infty} A_n = (\bigcup_{n=1}^{\infty} A_n^c)^c$. Each $A_n^c \in \mathcal{F}$ by (ii), so $\bigcup A_n^c \in \mathcal{F}$ by (iii), and $(\bigcup A_n^c)^c \in \mathcal{F}$ by (ii).

(c) $A \setminus B = A \cap B^c$. Since $B^c \in \mathcal{F}$, and $A \cap B^c = (A^c \cup B)^c \in \mathcal{F}$ by (ii) and (iii). For symmetric difference: $A \triangle B = (A \setminus B) \cup (B \setminus A)$.

(d) For finite unions $A_1 \cup \cdots \cup A_n$, set $A_k = \emptyset$ for $k > n$ in (iii). Finite intersections follow from (b) similarly. □

Example 2.1 (Extreme σ-Algebras)**.** For any Ω:

- The *trivial σ-algebra* $\{\emptyset, \Omega\}$ is the smallest σ-algebra on Ω.
- The *power set* $\mathcal{P}(\Omega) = \{A : A \subseteq \Omega\}$ is the largest.

Every σ-algebra on Ω lies between these two extremes: $\{\emptyset, \Omega\} \subseteq \mathcal{F} \subseteq \mathcal{P}(\Omega)$.

Example 2.2 (Partition-Generated σ-Algebra)**.** Let $\Omega = \{1, 2, 3, 4\}$ and consider the partition $\{\{1, 2\}, \{3, 4\}\}$. The σ-algebra generated by this partition is:

$$\mathcal{F} = \big\{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 2, 3, 4\}\big\}. \tag{1}$$

This has 4 elements. In general, a partition of Ω into k blocks generates a σ-algebra with 2^k elements (all possible unions of blocks).

Connection to RL

In a finite MDP with state space $\mathcal{S} = \{s_1, \ldots, s_n\}$, the "information contained in knowing S_t" is the σ-algebra generated by S_t: $\sigma(S_t) = \sigma\big(\{S_t = s_1\}, \ldots, \{S_t = s_n\}\big)$. This is a partition-generated σ-algebra with 2^n events. Each event $A \in \sigma(S_t)$ is a union of events $\{S_t = s_i\}$, so knowing which event occurred tells you exactly which state you are in.

2.2 Generated σ-Algebras

Given an arbitrary collection of subsets, we want the "smallest" σ-algebra containing them. This is essential because we often specify a few key events (or random variables) and need to derive the full σ-algebra they generate.

Lemma 2.2 (Intersection of σ-Algebras)**.** *If $\{\mathcal{F}_\alpha\}_{\alpha \in I}$ is any (possibly uncountable) collection of σ-algebras on Ω, then $\bigcap_{\alpha \in I} \mathcal{F}_\alpha$ is also a σ-algebra on Ω.*

Proof. We verify the three axioms.

(i) $\Omega \in \mathcal{F}_\alpha$ for all α, so $\Omega \in \bigcap \mathcal{F}_\alpha$.

(ii) If $A \in \bigcap \mathcal{F}_\alpha$, then $A \in \mathcal{F}_\alpha$ for all α, so $A^c \in \mathcal{F}_\alpha$ for all α, so $A^c \in \bigcap \mathcal{F}_\alpha$.

(iii) If $A_1, A_2, \ldots \in \bigcap \mathcal{F}_\alpha$, then for each α, all $A_n \in \mathcal{F}_\alpha$, so $\bigcup A_n \in \mathcal{F}_\alpha$. Hence $\bigcup A_n \in \bigcap \mathcal{F}_\alpha$. □

Definition 2.2 (Generated σ-Algebra)**.** For any collection $\mathcal{C} \subseteq \mathcal{P}(\Omega)$, the σ-algebra *generated by* $\mathcal{C}$ is:

$$\sigma(\mathcal{C}) := \bigcap \big\{\mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra on } \Omega,\ \mathcal{C} \subseteq \mathcal{G}\big\}. \tag{2}$$

Remark 2.2. This is well defined by Lemma 2.2: the intersection of σ-algebras is a σ-algebra, and the collection is non-empty (it includes $\mathcal{P}(\Omega)$). By construction, $\sigma(\mathcal{C})$ is the *smallest* σ-algebra containing $\mathcal{C}$: it contains $\mathcal{C}$, and any σ-algebra containing $\mathcal{C}$ must contain $\sigma(\mathcal{C})$.

Definition 2.3 (Borel σ-Algebra)**.** The *Borel σ-algebra* on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, is the σ-algebra generated by the open sets of $\mathbb{R}$:

$$\mathcal{B}(\mathbb{R}) := \sigma\big(\{U \subseteq \mathbb{R} : U \text{ is open}\}\big). \tag{3}$$

Elements of $\mathcal{B}(\mathbb{R})$ are called *Borel sets*.

Proposition 2.3 (Equivalent Generators of $\mathcal{B}(\mathbb{R})$)**.** *The following collections all generate the same σ-algebra $\mathcal{B}(\mathbb{R})$:*

(a) Open sets.
(b) Closed sets.
(c) Open intervals (a, b) with $a, b \in \mathbb{Q}$.
(d) Half-lines $(-\infty, x]$ with $x \in \mathbb{R}$.
(e) Half-lines $(-\infty, x)$ with $x \in \mathbb{Q}$.

Proof. We show that each collection generates a σ-algebra containing all the others.

(a) $\Leftrightarrow$ (b): Every closed set is the complement of an open set and vice versa. Since σ-algebras are closed under complement, $\sigma(\text{open sets}) = \sigma(\text{closed sets})$.

(c) $\Rightarrow$ (a): Every open set $U \subseteq \mathbb{R}$ is a countable union of open intervals with rational endpoints (since $\mathbb{Q}$ is dense: for each $x \in U$, find rational $a_x < x < b_x$ with $(a_x, b_x) \subseteq U$, and $U = \bigcup_{x \in U}(a_x, b_x)$; by density and separability, this is a countable union). So $\sigma(\text{rational open intervals}) \supseteq$ open sets.

(a) $\Rightarrow$ (c): Trivially, since every open interval with rational endpoints is open.

(d) $\Rightarrow$ (a): Note $(a, b) = (-\infty, b) \setminus (-\infty, a]$ and $(-\infty, b) = \bigcup_{n=1}^{\infty}(-\infty, b - 1/n]$. So open intervals are in $\sigma(\{(-\infty, x] : x \in \mathbb{R}\})$, and by (c) $\Rightarrow$ (a), we get all open sets.

(a) $\Rightarrow$ (d): $(-\infty, x] = \mathbb{R} \setminus (x, \infty)$, and (x, ∞) is open, so $(-\infty, x] \in \mathcal{B}(\mathbb{R})$.

(d) $\Leftrightarrow$ (e): $(-\infty, x] = \bigcap_{n=1}^{\infty}(-\infty, x + 1/n)$ (using rationals $x + 1/n$ for a sequence approaching from the right, taking $q_n \in \mathbb{Q}$ with $x < q_n < x + 1/n$). Conversely, $(-\infty, x) = \bigcup_{n=1}^{\infty}(-\infty, x - 1/n]$. □

Remark 2.3. Proposition 2.3(d) is particularly important. The characterization via half-lines $(-\infty, x]$ means that to check whether a function $f : \Omega \to \mathbb{R}$ is measurable, it suffices to verify $f^{-1}((-\infty, x]) = \{\omega : f(\omega) \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$. This is the standard criterion used for random variables.

2.3 The π–λ Theorem

The π–λ theorem (Dynkin's theorem) is one of the most useful tools in measure theory. It lets us prove that two measures agree on an entire σ-algebra by checking agreement on a much smaller collection.

Definition 2.4 (π-System)**.** A collection $\mathcal{P} \subseteq \mathcal{P}(\Omega)$ is a *π-system* if it is closed under finite intersection: $A, B \in \mathcal{P}$ implies $A \cap B \in \mathcal{P}$.

Definition 2.5 (λ-System (Dynkin System))**.** A collection $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ is a *λ-system* if:

(i) $\Omega \in \mathcal{L}$.
(ii) If $A, B \in \mathcal{L}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{L}$. (closure under proper difference)
(iii) If $A_1 \subseteq A_2 \subseteq \cdots$ are in $\mathcal{L}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{L}$. (closure under increasing union)

Remark 2.4. A σ-algebra is both a π-system and a λ-system. The power of the π–λ theorem is that the converse also holds: a collection that is both a π-system *and* a λ-system is a σ-algebra.

Lemma 2.4 ($\pi + \lambda = \sigma$)**.** *If $\mathcal{F}$ is both a π-system and a λ-system, then $\mathcal{F}$ is a σ-algebra.*

Proof. We verify the three σ-algebra axioms.

(i) $\Omega \in \mathcal{F}$ by the λ-system property.

(ii) If $A \in \mathcal{F}$, then $A \subseteq \Omega$ and both are in $\mathcal{F}$, so $A^c = \Omega \setminus A \in \mathcal{F}$ by the proper difference property.

(iii) Let $A_1, A_2, \ldots \in \mathcal{F}$. Define $B_n = \bigcup_{k=1}^{n} A_k$. We need to show $B_n \in \mathcal{F}$ for each n (then $\bigcup A_n = \bigcup B_n$ is an increasing union in $\mathcal{F}$).

For $n = 1$: $B_1 = A_1 \in \mathcal{F}$. For the inductive step: assuming $B_n \in \mathcal{F}$, we write $B_{n+1} = B_n \cup A_{n+1}$. Now:

$$B_n \cup A_{n+1} = (B_n^c \cap A_{n+1}^c)^c. \tag{4}$$

Since $\mathcal{F}$ is closed under complement (shown above) and intersection (π-system), $B_n^c \cap A_{n+1}^c \in \mathcal{F}$, hence $(B_n^c \cap A_{n+1}^c)^c \in \mathcal{F}$.

So B_n is an increasing sequence in $\mathcal{F}$, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n \in \mathcal{F}$ by the λ-system property. □

Dynkin's π–λ Theorem

Theorem 2.5 (π–λ Theorem)**.** *If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system with $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.*

Proof. Let $\ell(\mathcal{P})$ denote the smallest λ-system containing $\mathcal{P}$ (this exists by the same intersection argument as for generated σ-algebras: $\mathcal{P}(\Omega)$ is a λ-system, and the intersection of λ-systems is a λ-system). Since $\mathcal{L}$ is a λ-system containing $\mathcal{P}$, we have $\ell(\mathcal{P}) \subseteq \mathcal{L}$.

It suffices to show $\ell(\mathcal{P})$ is a σ-algebra (for then $\sigma(\mathcal{P}) \subseteq \ell(\mathcal{P}) \subseteq \mathcal{L}$). By Lemma 2.4, it suffices to show $\ell(\mathcal{P})$ is a π-system.

Step 1: For each $B \in \ell(\mathcal{P})$, define $\mathcal{G}_B = \{A \in \ell(\mathcal{P}) : A \cap B \in \ell(\mathcal{P})\}$. We claim $\mathcal{G}_B$ is a λ-system:

- $\Omega \in \mathcal{G}_B$ since $\Omega \cap B = B \in \ell(\mathcal{P})$.
- If $A_1 \subseteq A_2$ with $A_1, A_2 \in \mathcal{G}_B$, then $(A_2 \setminus A_1) \cap B = (A_2 \cap B) \setminus (A_1 \cap B)$, and $A_1 \cap B \subseteq A_2 \cap B$ with both in $\ell(\mathcal{P})$, so $(A_2 \setminus A_1) \cap B \in \ell(\mathcal{P})$.
- If $A_n \uparrow$, then $A_n \cap B \uparrow$, so $(\bigcup A_n) \cap B = \bigcup (A_n \cap B) \in \ell(\mathcal{P})$.

Step 2: For $B \in \mathcal{P}$: since $\mathcal{P}$ is a π-system, $A \cap B \in \mathcal{P} \subseteq \ell(\mathcal{P})$ for all $A \in \mathcal{P}$. So $\mathcal{P} \subseteq \mathcal{G}_B$. Since $\mathcal{G}_B$ is a λ-system, $\ell(\mathcal{P}) \subseteq \mathcal{G}_B$. This means: for all $B \in \mathcal{P}$ and $A \in \ell(\mathcal{P})$, $A \cap B \in \ell(\mathcal{P})$.

Step 3: Now fix $A \in \ell(\mathcal{P})$ and consider $\mathcal{G}_A = \{B \in \ell(\mathcal{P}) : A \cap B \in \ell(\mathcal{P})\}$. By Step 2, $\mathcal{P} \subseteq \mathcal{G}_A$ (since for $B \in \mathcal{P}$, $A \cap B \in \ell(\mathcal{P})$ by Step 2). Since $\mathcal{G}_A$ is a λ-system (by Step 1 with A and B swapped), $\ell(\mathcal{P}) \subseteq \mathcal{G}_A$.

This means: for all $A, B \in \ell(\mathcal{P})$, $A \cap B \in \ell(\mathcal{P})$. So $\ell(\mathcal{P})$ is a π-system. By Lemma 2.4, $\ell(\mathcal{P})$ is a σ-algebra, so $\sigma(\mathcal{P}) \subseteq \ell(\mathcal{P}) \subseteq \mathcal{L}$. □

Corollary 2.6 (Uniqueness of Measures)**.** *If two probability measures $\mathbb{P}_1$ and $\mathbb{P}_2$ on $(\Omega, \sigma(\mathcal{P}))$ agree on a π-system $\mathcal{P}$, then $\mathbb{P}_1 = \mathbb{P}_2$ on $\sigma(\mathcal{P})$.*

Proof. The collection $\mathcal{L} = \{A \in \sigma(\mathcal{P}) : \mathbb{P}_1(A) = \mathbb{P}_2(A)\}$ is a λ-system containing $\mathcal{P}$. By the π–λ theorem, $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.

We verify $\mathcal{L}$ is a λ-system:

- $\Omega \in \mathcal{L}$ since $\mathbb{P}_1(\Omega) = \mathbb{P}_2(\Omega) = 1$.
- If $A \subseteq B$ with $A, B \in \mathcal{L}$, then $\mathbb{P}_i(B \setminus A) = \mathbb{P}_i(B) - \mathbb{P}_i(A)$ for $i = 1, 2$, and these are equal.
- If $A_n \uparrow A$ with $A_n \in \mathcal{L}$, then $\mathbb{P}_i(A) = \lim \mathbb{P}_i(A_n)$ by continuity from below, and these limits are equal. □

Connection to RL

Corollary 2.6 is why the distribution of a random variable is uniquely determined by its CDF $F_X(x) = \mathbb{P}(X \leq x)$: the half-lines $(-\infty, x]$ form a π-system generating $\mathcal{B}(\mathbb{R})$, so any two measures agreeing on all $(-\infty, x]$ must agree on all Borel sets. In RL, this means the distribution of the return G_t is fully characterized by its CDF — there is no "hidden" probabilistic information.

3 Measures

3.1 Definition and Fundamental Properties

Definition 3.1 (Measure)**.** A *measure* on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

(i) $\mu(\emptyset) = 0$.

(ii) **Countable additivity:** If $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then:

$$\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{5}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*. If $\mu(\Omega) = 1$, then μ is a *probability measure* and we write $\mathbb{P}$ instead of μ.

Example 3.1 (Standard Measures)**.** (a) **Counting measure:** $\mu(A) = |A|$ (number of elements) on any countable Ω.

(b) **Dirac measure:** $\delta_x(A) = \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$.

(c) **Lebesgue measure:** $\lambda((a, b]) = b - a$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. (Existence proved in Section 4.)

(d) **Probability on a finite set:** $\mathbb{P}(\{s\}) = p(s)$ with $\sum_{s \in \mathcal{S}} p(s) = 1$.

Theorem 3.1 (Properties of Measures)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For $A, B, A_n \in \mathcal{F}$:*

(a) ***Monotonicity:*** $A \subseteq B \implies \mu(A) \leq \mu(B)$.

(b) ***Subtractivity:*** $A \subseteq B$ *and* $\mu(A) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$.

(c) ***Sub-additivity:*** $\mu(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$.

(d) ***Continuity from below:*** $A_n \uparrow A \implies \mu(A) = \lim_n \mu(A_n)$.

(e) ***Continuity from above:*** $A_n \downarrow A$ *and* $\mu(A_1) < \infty \implies \mu(A) = \lim_n \mu(A_n)$.

Proof. **(a)** $B = A \sqcup (B \setminus A)$, so $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(b) From $\mu(B) = \mu(A) + \mu(B \setminus A)$ and $\mu(A) < \infty$, subtract $\mu(A)$.

(c) Define $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. The B_n are disjoint, $\bigcup B_n = \bigcup A_n$, and $B_n \subseteq A_n$. So: $\mu(\bigcup A_n) = \mu(\bigsqcup B_n) = \sum \mu(B_n) \leq \sum \mu(A_n)$.

(d) Let $A_0 = \emptyset$. The sets $C_n = A_n \setminus A_{n-1}$ are disjoint and $A_n = \bigsqcup_{k=1}^{n} C_k$. Also $A = \bigsqcup_{k=1}^{\infty} C_k$. By countable additivity:

$$\mu(A) = \sum_{k=1}^{\infty} \mu(C_k) = \lim_{n \to \infty} \sum_{k=1}^{n} \mu(C_k) = \lim_{n \to \infty} \mu(A_n). \tag{6}$$

(e) Define $B_n = A_1 \setminus A_n$. Then $B_n \uparrow A_1 \setminus A$. By (d): $\mu(A_1 \setminus A) = \lim \mu(B_n) = \lim(\mu(A_1) - \mu(A_n))$, using (b) (valid since $\mu(A_n) \leq \mu(A_1) < \infty$). So $\mu(A_1) - \mu(A) = \mu(A_1) - \lim \mu(A_n)$, giving $\mu(A) = \lim \mu(A_n)$. □

Finiteness Condition in (e)

The condition $\mu(A_1) < \infty$ in continuity from above is essential. Consider counting measure on $\mathbb{N}$ with $A_n = \{n, n+1, n+2, \ldots\}$. Then $A_n \downarrow \emptyset$ but $\mu(A_n) = \infty$ for all n, while $\mu(\emptyset) = 0$. For probability measures, this condition is always satisfied since $\mathbb{P}(A_1) \leq 1 < \infty$.

3.2 Null Sets and Complete Measures

Definition 3.2 (Null Set, Complete Measure)**.** A set $N \in \mathcal{F}$ is a *μ-null set* (or *null set*) if $\mu(N) = 0$. A measure μ is *complete* if whenever N is μ-null and $A \subseteq N$, then $A \in \mathcal{F}$ (and hence $\mu(A) = 0$).

Proposition 3.2 (Completion of a Measure)**.** *Every measure space $(\Omega, \mathcal{F}, \mu)$ can be completed: there exists a smallest σ-algebra $\overline{\mathcal{F}} \supseteq \mathcal{F}$ and a measure $\bar{\mu}$ extending μ such that $(\Omega, \overline{\mathcal{F}}, \bar{\mu})$ is complete. Explicitly:*

$$\overline{\mathcal{F}} = \{A \cup N' : A \in \mathcal{F},\ N' \subseteq N \text{ for some } \mu\text{-null } N \in \mathcal{F}\}, \tag{7}$$

and $\bar{\mu}(A \cup N') = \mu(A)$.

Proof. We verify that $\overline{\mathcal{F}}$ is a σ-algebra. The key step is closure under complement: if $E = A \cup N'$ with $N' \subseteq N$ and $\mu(N) = 0$, then $E^c = A^c \cap (N')^c = (A^c \cap N^c) \cup (A^c \cap N \cap (N')^c)$. The first part is in $\mathcal{F}$, and the second is a subset of N (a null set), so $E^c \in \overline{\mathcal{F}}$.

For well-definedness of $\bar{\mu}$: if $A_1 \cup N_1' = A_2 \cup N_2'$, then $A_1 \triangle A_2 \subseteq N_1 \cup N_2$ (a null set), so $\mu(A_1 \triangle A_2) = 0$, hence $\mu(A_1) = \mu(A_2)$. □

4 Construction of Measures: Carathéodory's Theorem

How do we actually *build* measures? We cannot simply assign $\mu(A)$ for every $A \in \mathcal{B}(\mathbb{R})$ individually — the Borel σ-algebra is uncountably large. Instead, we specify μ on a small, manageable collection and extend it.

4.1 Algebras and Pre-Measures

Definition 4.1 (Algebra of Sets)**.** A collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an *algebra* (or *field*) if:

(i) $\Omega \in \mathcal{A}$.
(ii) $A \in \mathcal{A} \implies A^c \in \mathcal{A}$.
(iii) $A, B \in \mathcal{A} \implies A \cup B \in \mathcal{A}$. (closed under *finite* union)

Remark 4.1. An algebra is like a σ-algebra but with only finite (not countable) closure. Every σ-algebra is an algebra, but not conversely.

Example 4.1. On $\mathbb{R}$, the collection $\mathcal{A}$ of all finite unions of half-open intervals $(a, b]$ (including $\emptyset$ and $\mathbb{R} = (-\infty, \infty)$) is an algebra but not a σ-algebra. For instance, $\bigcup_{n=1}^{\infty}(0, 1 - 1/n] = (0, 1)$ is not a finite union of half-open intervals, so $(0, 1) \notin \mathcal{A}$.

Definition 4.2 (Pre-Measure)**.** A *pre-measure* on an algebra $\mathcal{A}$ is a function $\mu_0 : \mathcal{A} \to [0, \infty]$ satisfying:

(i) $\mu_0(\emptyset) = 0$.
(ii) If $A_1, A_2, \ldots \in \mathcal{A}$ are disjoint and $\bigsqcup_{n=1}^{\infty} A_n \in \mathcal{A}$, then $\mu_0(\bigsqcup A_n) = \sum \mu_0(A_n)$.

4.2 Outer Measures

Definition 4.3 (Outer Measure)**.** An *outer measure* on Ω is a function $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ satisfying:

(i) $\mu^*(\emptyset) = 0$.
(ii) **Monotonicity:** $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$.
(iii) **Countable sub-additivity:** $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Proposition 4.1 (Pre-Measure Induces Outer Measure)**.** *Given a pre-measure μ_0 on an algebra $\mathcal{A}$, define for any $E \subseteq \Omega$:*

$$\mu^*(E) = \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : A_n \in \mathcal{A},\ E \subseteq \bigcup_{n=1}^{\infty} A_n\right\}. \tag{8}$$

Then μ^* is an outer measure, and $\mu^*(A) = \mu_0(A)$ for all $A \in \mathcal{A}$.

Proof. *(i)* $\mu^*(\emptyset) = 0$: cover $\emptyset$ by $A_n = \emptyset$ for all n.

(ii) If $A \subseteq B$, every cover of B is also a cover of A, so the infimum over covers of A is at most the infimum over covers of B.

(iii) Let $E = \bigcup_n E_n$. For each $\varepsilon > 0$ and each n, choose a cover $\{A_{n,k}\}_k \subseteq \mathcal{A}$ with $E_n \subseteq \bigcup_k A_{n,k}$ and $\sum_k \mu_0(A_{n,k}) \leq \mu^*(E_n) + \varepsilon/2^n$. Then $\{A_{n,k}\}_{n,k}$ covers E, so:

$$\mu^*(E) \leq \sum_n \sum_k \mu_0(A_{n,k}) \leq \sum_n (\mu^*(E_n) + \varepsilon/2^n) = \sum_n \mu^*(E_n) + \varepsilon. \tag{9}$$

Since ε is arbitrary, $\mu^*(E) \leq \sum_n \mu^*(E_n)$.

μ^* extends μ_0: For $A \in \mathcal{A}$, the single-set cover gives $\mu^*(A) \leq \mu_0(A)$. For the reverse: if $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}$, define $B_n = A \cap A_n \cap (\bigcup_{k<n} A_k)^c \in \mathcal{A}$ (since $\mathcal{A}$ is an algebra). Then $A = \bigsqcup B_n$ and $B_n \subseteq A_n$. By the pre-measure properties: $\mu_0(A) = \sum \mu_0(B_n) \leq \sum \mu_0(A_n)$. Taking the infimum: $\mu_0(A) \leq \mu^*(A)$. □

4.3 Carathéodory's Extension Theorem

Definition 4.4 (Carathéodory Measurability)**.** Given an outer measure μ^* on Ω, a set $A \subseteq \Omega$ is *μ^*-measurable* (or *Carathéodory measurable*) if for every $E \subseteq \Omega$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c). \tag{10}$$

Remark 4.2. By sub-additivity, $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ always holds. So the content of (10) is the reverse inequality: A "splits" every test set E additively. This is a strong condition — it must hold for *all* $E \subseteq \Omega$, not just sets in some σ-algebra.

Carathéodory's Extension Theorem

Theorem 4.2 (Carathéodory)**.** *Let μ^* be an outer measure on Ω. Then:*

(a) The collection $\mathcal{M} = \{A \subseteq \Omega : A$ is μ^*-measurable$\}$ is a σ-algebra.

(b) The restriction $\mu = \mu^*|_{\mathcal{M}}$ is a complete measure on $(\Omega, \mathcal{M})$.

Consequently, if μ_0 is a pre-measure on an algebra $\mathcal{A}$ and μ^* is the induced outer measure (8), then $\mathcal{A} \subseteq \mathcal{M}$ and $\mu|_{\mathcal{A}} = \mu_0$. If additionally μ_0 is σ-finite, then μ is the unique extension of μ_0 to $\sigma(\mathcal{A})$.

Proof. **Part (a): $\mathcal{M}$ is a σ-algebra.**

Step 1: $\Omega \in \mathcal{M}$. For any E: $\mu^*(E \cap \Omega) + \mu^*(E \cap \emptyset) = \mu^*(E) + 0 = \mu^*(E)$. ✓

Step 2: Closure under complement. The condition (10) is symmetric in A and A^c, so $A \in \mathcal{M} \Leftrightarrow A^c \in \mathcal{M}$. ✓

Step 3: Closure under finite union. Let $A, B \in \mathcal{M}$. For any test set E:

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && (A \text{ measurable})\\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && (B \text{ measurable, applied to } E \cap A^c)\end{aligned}$$

We need $\mu^*(E) = \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$.

Note $(A \cup B)^c = A^c \cap B^c$, so the third term is correct. For the first: $E \cap (A \cup B) = (E \cap A) \cup (E \cap A^c \cap B)$, so by sub-additivity: $\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B)$.

Combining: $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. The reverse holds by sub-additivity. So $A \cup B \in \mathcal{M}$. ✓

Step 4: Closure under countable disjoint union. Let $A_1, A_2, \ldots \in \mathcal{M}$ be disjoint. Set $S_n = \bigsqcup_{k=1}^n A_k$ and $S = \bigsqcup_{k=1}^\infty A_k$.

By induction (using Step 3 and closure under complement, hence closure under intersection), $S_n \in \mathcal{M}$, and for any E:

$$\mu^*(E \cap S_n) = \sum_{k=1}^n \mu^*(E \cap A_k). \tag{11}$$

This is proved by induction: $\mu^*(E \cap S_n) = \mu^*(E \cap S_n \cap A_n) + \mu^*(E \cap S_n \cap A_n^c) = \mu^*(E \cap A_n) + \mu^*(E \cap S_{n-1})$.

Since $S_n \in \mathcal{M}$: $\mu^*(E) = \mu^*(E \cap S_n) + \mu^*(E \cap S_n^c)$. Since $S_n \subseteq S$, $S^c \subseteq S_n^c$, so $\mu^*(E \cap S_n^c) \geq \mu^*(E \cap S^c)$. Using (11):

$$\mu^*(E) \geq \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*(E \cap S^c). \tag{12}$$

Letting $n \to \infty$:

$$\mu^*(E) \geq \sum_{k=1}^\infty \mu^*(E \cap A_k) + \mu^*(E \cap S^c) \geq \mu^*(E \cap S) + \mu^*(E \cap S^c), \tag{13}$$

where the last step uses sub-additivity $\mu^*(E \cap S) \leq \sum \mu^*(E \cap A_k)$. The reverse inequality holds by sub-additivity. So $S \in \mathcal{M}$. ✓

Step 5: General countable unions. For arbitrary (not necessarily disjoint) $A_n \in \mathcal{M}$, write $\bigcup A_n$ as a disjoint union using $B_n = A_n \setminus \bigcup_{k<n} A_k \in \mathcal{M}$, and apply Step 4.

Part (b): μ is a complete measure.

Countable additivity follows from (11) with $E = S$: $\mu(S) = \mu^*(S) = \sum \mu^*(A_k) = \sum \mu(A_k)$.

Completeness: if $\mu^*(N) = 0$ and $A \subseteq N$, then for any E: $\mu^*(E \cap A) \leq \mu^*(A) \leq \mu^*(N) = 0$, so $\mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) \leq \mu^*(E)$. The reverse holds by monotonicity ($E \cap A^c \subseteq E$). Wait, actually we need $\mu^*(E \cap A^c) \geq \mu^*(E) - \mu^*(E \cap A) = \mu^*(E)$ and $\mu^*(E \cap A^c) \leq \mu^*(E)$. These give equality. So $A \in \mathcal{M}$.

Uniqueness under σ-finiteness: follows from Corollary 2.6 applied to the π-system $\mathcal{A}$, after reducing to the finite-measure case using the σ-finite decomposition. □

4.4 Application: Lebesgue Measure

Theorem 4.3 (Existence of Lebesgue Measure)**.** *There exists a unique measure λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\lambda((a, b]) = b - a$ for all $a < b$. This measure is σ-finite and translation-invariant: $\lambda(A + x) = \lambda(A)$ for all $A \in \mathcal{B}(\mathbb{R})$ and $x \in \mathbb{R}$.*

Proof sketch. Let $\mathcal{A}$ be the algebra of finite disjoint unions of half-open intervals $(a_i, b_i]$. Define $\mu_0(\bigsqcup_i (a_i, b_i]) = \sum_i (b_i - a_i)$. One verifies μ_0 is a pre-measure on $\mathcal{A}$ (the key step is showing countable additivity, which uses the Heine–Borel theorem: if $(a, b]$ is covered by countably many $(a_n, b_n]$, then $b - a \leq \sum (b_n - a_n)$).

By Carathéodory's theorem (Theorem 4.2), μ_0 extends uniquely to $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$ since μ_0 is σ-finite ($\mathbb{R} = \bigcup_{n=1}^\infty (-n, n]$ with $\mu_0((-n, n]) = 2n < \infty$). □

Theorem 4.4 (Non-Measurable Sets (Vitali))**.** *There exists a set $V \subseteq [0, 1]$ that is not Lebesgue measurable.*

Proof. Define an equivalence relation on $[0,1)$: $x \sim y$ iff $x - y \in \mathbb{Q}$. By the Axiom of Choice, select one representative from each equivalence class; call this set V.

For $q \in \mathbb{Q} \cap [0,1)$, define $V_q = \{x + q \mod 1 : x \in V\}$ (shift V by q modulo 1). The V_q are disjoint (if $v_1 + q_1 \equiv v_2 + q_2$ then $v_1 - v_2 \in \mathbb{Q}$, so $v_1 \sim v_2$, so $v_1 = v_2$ and $q_1 = q_2$). Also $[0,1) = \bigsqcup_{q \in \mathbb{Q} \cap [0,1)} V_q$ (every x is equivalent to some $v \in V$, and $x = v + q$ for some rational q).

If V were measurable, then each V_q would have the same measure (by translation invariance of Lebesgue measure modulo 1). Call this c.

By countable additivity: $1 = \lambda([0,1)) = \sum_q c$. But $\sum c$ is either 0 (if $c = 0$) or ∞ (if $c > 0$) — never 1. Contradiction. □

Remark 4.3. Vitali's theorem shows why we cannot define Lebesgue measure on $\mathcal{P}(\mathbb{R})$: not all subsets of $\mathbb{R}$ are measurable. This is the fundamental reason for restricting to σ-algebras. In probability, this subtlety is rarely encountered (finite MDPs use $\mathcal{P}(\mathcal{S})$), but it justifies the formalism and is important for continuous state/action spaces.

5 Measurable Functions

5.1 Definition and Characterization

Definition 5.1 (Measurable Function)**.** Let $(\Omega, \mathcal{F})$ and $(S, \mathcal{S})$ be measurable spaces. A function $f : \Omega \to S$ is $(\mathcal{F}, \mathcal{S})$-*measurable* (or simply *measurable*) if:

$$f^{-1}(B) := \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}. \tag{14}$$

When $S = \mathbb{R}$ and $\mathcal{S} = \mathcal{B}(\mathbb{R})$, we say f is *Borel measurable* (or just *measurable*).

Theorem 5.1 (Generator Criterion for Measurability)**.** *Let $\mathcal{S} = \sigma(\mathcal{C})$ for some collection $\mathcal{C}$. Then $f : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ is measurable if and only if:*

$$f^{-1}(C) \in \mathcal{F} \quad \textit{for all } C \in \mathcal{C}. \tag{15}$$

Proof. ($\Rightarrow$) Trivial: $\mathcal{C} \subseteq \mathcal{S}$.

($\Leftarrow$) Define $\mathcal{G} = \{B \in \mathcal{S} : f^{-1}(B) \in \mathcal{F}\}$. We show $\mathcal{G}$ is a σ-algebra:

- $S \in \mathcal{G}$: $f^{-1}(S) = \Omega \in \mathcal{F}$.
- $B \in \mathcal{G} \implies B^c \in \mathcal{G}$: $f^{-1}(B^c) = (f^{-1}(B))^c \in \mathcal{F}$.
- $B_n \in \mathcal{G} \implies \bigcup B_n \in \mathcal{G}$: $f^{-1}(\bigcup B_n) = \bigcup f^{-1}(B_n) \in \mathcal{F}$.

By hypothesis, $\mathcal{C} \subseteq \mathcal{G}$. Since $\mathcal{G}$ is a σ-algebra, $\sigma(\mathcal{C}) = \mathcal{S} \subseteq \mathcal{G}$, so $\mathcal{G} = \mathcal{S}$. □

Corollary 5.2 (Standard Criterion for Real-Valued Measurability)**.** *$f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable if and only if $\{f \le x\} := \{\omega : f(\omega) \le x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$.*

Proof. Apply Theorem 5.1 with $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$, which generates $\mathcal{B}(\mathbb{R})$ by Proposition 2.3(d). □

Connection to RL

A *random variable* is precisely a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In an MDP, the state S_t, action A_t, and reward R_{t+1} are all random variables. The value function $V^\pi(s) = \mathbb{E}[G_t \mid S_t = s]$ is a measurable function of s. Corollary 5.2 says that being a random variable means precisely that events like $\{X \le x\}$ are "observable" (in $\mathcal{F}$) — which is why we can compute their probabilities.

5.2 Algebra of Measurable Functions

Theorem 5.3 (Closure Properties)**.** *Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and $c \in \mathbb{R}$. Then the following are measurable:*

(a) $cf, \quad f+g, \quad fg, \quad |f|, \quad \max(f,g), \quad \min(f,g)$.
(b) f/g *on* $\{g \neq 0\}$.
(c) $f^+ = \max(f, 0)$ *and* $f^- = \max(-f, 0)$.

Proof. We prove $f+g$ is measurable; the others are similar.

Key idea: $\{f+g<c\} = \bigcup_{q\in\mathbb{Q}}(\{f<q\} \cap \{g<c-q\})$.

Why? If $f(\omega)+g(\omega)<c$, then $f(\omega)<c-g(\omega)$. By density of $\mathbb{Q}$, there exists $q \in \mathbb{Q}$ with $f(\omega)<q<c-g(\omega)$, so $\omega \in \{f<q\} \cap \{g<c-q\}$.

Conversely, if $f(\omega)<q$ and $g(\omega)<c-q$, then $f(\omega)+g(\omega)<c$.

Since each $\{f<q\}$ and $\{g<c-q\}$ is in $\mathcal{F}$ (measurability of f and g), and the union is countable (over $q \in \mathbb{Q}$), we conclude $\{f+g<c\} \in \mathcal{F}$.

Since $\{f+g<c\} \in \mathcal{F}$ for all c and $\{f+g \leq c\} = \bigcap_n \{f+g<c+1/n\}$, we get $\{f+g \leq c\} \in \mathcal{F}$. □

Theorem 5.4 (Limits of Measurable Functions)**.** *If $f_1, f_2, \ldots$ are measurable, then the following are measurable:*

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_n f_n, \quad \liminf_n f_n. \tag{16}$$

In particular, if $f_n \to f$ pointwise (or a.s.), then f is measurable.

Proof. $\{\sup_n f_n \leq c\} = \bigcap_{n=1}^{\infty}\{f_n \leq c\} \in \mathcal{F}$ (countable intersection of measurable sets).

$\inf_n f_n = -\sup_n(-f_n)$ is measurable.

$\limsup_n f_n = \inf_n \sup_{k\geq n} f_k$ and $\liminf_n f_n = \sup_n \inf_{k\geq n} f_k$ are measurable by composing the above.

If $f_n \to f$ pointwise, then $f = \lim_n f_n = \limsup_n f_n = \liminf_n f_n$. □

5.3 Simple Functions and Approximation

Definition 5.2 (Simple Function)**.** A *simple function* is a measurable function $\varphi : \Omega \to \mathbb{R}$ taking finitely many values. Every simple function can be written as:

$$\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}, \tag{17}$$

where $a_1, \ldots, a_n \in \mathbb{R}$ are distinct, $A_i = \varphi^{-1}(\{a_i\}) \in \mathcal{F}$, and $\Omega = \bigsqcup_{i=1}^{n} A_i$. This is the *standard representation.*

Simple Function Approximation

Theorem 5.5 (Approximation Theorem)**.** *Let $f : (\Omega, \mathcal{F}) \to [0, \infty]$ be measurable. Then there exist simple functions $0 \leq \varphi_1 \leq \varphi_2 \leq \cdots$ with $\varphi_n \uparrow f$ pointwise. If f is bounded, the convergence is uniform.*

Proof. For each $n \geq 1$, define:

$$\varphi_n(\omega) = \begin{cases} \frac{k-1}{2^n} & \text{if } \frac{k-1}{2^n} \leq f(\omega) < \frac{k}{2^n}, \quad k = 1, 2, \ldots, n \cdot 2^n, \\ n & \text{if } f(\omega) \geq n. \end{cases} \tag{18}$$

In other words, φ_n rounds f down to the nearest multiple of $1/2^n$, capped at n.

Measurability: φ_n takes finitely many values, and each level set $\varphi_n^{-1}(\{k/2^n\}) = \{k/2^n \le f < (k+1)/2^n\} \in \mathcal{F}$.

Monotonicity: φ_{n+1} uses a finer grid ($1/2^{n+1}$ vs $1/2^n$) and a higher cap ($n+1$ vs n), so $\varphi_{n+1} \ge \varphi_n$.

Convergence: If $f(\omega) < \infty$, then for $n > f(\omega)$: $0 \le f(\omega) - \varphi_n(\omega) < 1/2^n \to 0$. If $f(\omega) = \infty$, then $\varphi_n(\omega) = n \to \infty = f(\omega)$.

Uniform convergence when bounded: If $f \le M$, then for $n > M$, $|f - \varphi_n| < 1/2^n$ everywhere. □

Connection to RL

The approximation theorem is the foundation of Lebesgue integration (next lesson). The Lebesgue integral of a non-negative function is defined as $\int f\,d\mu = \sup\{\int \varphi\,d\mu : \varphi \text{ simple}, \varphi \le f\}$, where simple function integrals are just weighted sums. In RL, this underlies the expectation $\mathbb{E}[G_t \mid S_t = s]$: the return G_t is approximated by simple functions, and the expectation is the limit of their integrals.

5.4 σ-Algebra Generated by a Random Variable

Definition 5.3 ($\sigma(X)$)**.** For a measurable function $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$, the σ-algebra *generated by* X is:

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{S}\} = \{\{X \in B\} : B \in \mathcal{S}\}. \tag{19}$$

This is the smallest σ-algebra on Ω making X measurable.

Theorem 5.6 (Doob–Dynkin Lemma)**.** *Let $X : \Omega \to S$ and $Y : \Omega \to \mathbb{R}$ be functions. Then Y is $\sigma(X)$-measurable if and only if $Y = g \circ X$ for some measurable $g : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.*

Proof. ($\Leftarrow$) If $Y = g(X)$, then $\{Y \le c\} = \{g(X) \le c\} = X^{-1}(g^{-1}((-\infty, c])) \in \sigma(X)$.

($\Rightarrow$) We prove this for the case $S = \mathbb{R}^d$ (which covers all RL applications).

Step 1: Simple functions. If $Y = \sum a_i \mathbf{1}_{A_i}$ with $A_i \in \sigma(X)$, then $A_i = X^{-1}(B_i)$ for some $B_i \in \mathcal{S}$. Define $g = \sum a_i \mathbf{1}_{B_i}$. Then $Y = g(X)$.

Step 2: Non-negative functions. By Theorem 5.5, $Y = \lim_n \varphi_n$ with φ_n simple and $\sigma(X)$-measurable. By Step 1, $\varphi_n = g_n(X)$. Define $g = \lim_n g_n$ (well-defined $\mathcal{S}$-a.e.). Then $Y = g(X)$.

Step 3: General. Write $Y = Y^+ - Y^-$ and apply Step 2 to each part. □

Connection to RL

The Doob–Dynkin lemma is the formal justification for the pull-out property: $\mathbb{E}[h(X) \cdot Y \mid X] = h(X) \cdot \mathbb{E}[Y \mid X]$. The function $h(X)$ is $\sigma(X)$-measurable (it's a function of X), so it's "known" given X and can be pulled out of the conditional expectation. This is used in every TD proof when extracting $V(S_t)$ from $\mathbb{E}[\delta_t \mid S_t]$.

6 From Measures to Probability

6.1 Probability Spaces

Definition 6.1 (Probability Space)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. The elements of $\mathcal{F}$ are called *events*, and $\mathbb{P}(A)$ is the *probability* of event A.

Definition 6.2 (Random Variable)**.** A *random variable* on $(\Omega, \mathcal{F}, \mathbb{P})$ is a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 6.3 (Distribution (Pushforward Measure))**.** The *distribution* (or *law*) of a random variable X is the probability measure μ_X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by:

$$\mu_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}). \tag{20}$$

The *cumulative distribution function* (CDF) is $F_X(x) = \mathbb{P}(X \leq x) = \mu_X((-\infty, x])$.

Proposition 6.1 (CDF Determines Distribution)**.** *The CDF F_X uniquely determines μ_X.*

Proof. The half-lines $\{(-\infty, x] : x \in \mathbb{R}\}$ form a π-system generating $\mathcal{B}(\mathbb{R})$. Two measures agreeing on a π-system agree on the generated σ-algebra (Corollary 2.6). Since $\mu_X((-\infty, x]) = F_X(x)$, the CDF determines μ_X on the π-system, hence on $\mathcal{B}(\mathbb{R})$. □

6.2 The MDP Probability Space

Example 6.1 (Probability Space for a Finite MDP)**.** Consider a finite MDP with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, and policy π. An infinite trajectory is:

$$\omega = (s_0, a_0, r_1, s_1, a_1, r_2, s_2, \ldots). \tag{21}$$

The probability space is:

- $\Omega = (\mathcal{S} \times \mathcal{A} \times \mathbb{R})^\infty$ — the set of all possible trajectories.
- $\mathcal{F} = \sigma(S_0, A_0, R_1, S_1, A_1, R_2, \ldots)$ — the σ-algebra generated by all coordinate projections.
- $\mathbb{P}_\pi$ — the probability measure determined by the initial distribution, policy, and transition kernel via the chain rule:

$$\mathbb{P}_\pi(S_0 = s_0, A_0 = a_0, S_1 = s_1, \ldots) = p_0(s_0) \prod_{t=0}^{\infty} \pi(a_t|s_t)\, P(s_{t+1}|s_t, a_t). \tag{22}$$

The filtration $\mathcal{F}_t = \sigma(S_0, A_0, R_1, \ldots, S_t)$ captures "information available at time t." A function Y is $\mathcal{F}_t$-measurable if and only if $Y = g(S_0, A_0, R_1, \ldots, S_t)$ for some Borel g (Doob–Dynkin).

7 Exercises

Exercise 7.1. Show that $|\mathcal{F}|$ (the number of elements in a finite σ-algebra on a finite set Ω) is always a power of 2. *Hint:* Show that every finite σ-algebra is generated by a partition of Ω.

Exercise 7.2. Prove that $\mathcal{B}(\mathbb{R})$ has the cardinality of the continuum ($|\mathcal{B}(\mathbb{R})| = |\mathbb{R}|$), while $|\mathcal{P}(\mathbb{R})| = 2^{|\mathbb{R}|} > |\mathbb{R}|$. Conclude that most subsets of $\mathbb{R}$ are not Borel sets.

Exercise 7.3. Let μ be a finite measure on $(\Omega, \mathcal{F})$ and $A_n \in \mathcal{F}$. Prove the *first Borel–Cantelli lemma*: if $\sum_n \mu(A_n) < \infty$, then $\mu(\limsup_n A_n) = 0$, where $\limsup_n A_n = \bigcap_{N=1}^\infty \bigcup_{n=N}^\infty A_n$. *Hint:* Use continuity from above and the union bound.

Exercise 7.4. Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable. Prove that $\{f < g\}$, $\{f \leq g\}$, $\{f = g\}$ are all in $\mathcal{F}$. *Hint:* $\{f < g\} = \bigcup_{q \in \mathbb{Q}} (\{f < q\} \cap \{q < g\})$.

Exercise 7.5. Let $\mathcal{A}$ be an algebra on Ω. A *monotone class* is a collection closed under increasing unions and decreasing intersections. Prove the *Monotone Class Theorem*: the smallest monotone class containing $\mathcal{A}$ equals $\sigma(\mathcal{A})$. *Hint:* Adapt the proof of the π–λ theorem.

Exercise 7.6. Construct a probability measure on $(\{0,1\}^\mathbb{N}, \mathcal{F})$ corresponding to an infinite sequence of fair coin flips, where $\mathcal{F}$ is generated by the cylinder sets. Verify that it satisfies countable additivity.

Exercise 7.7 (RL Application)**.** Consider an MDP with $\mathcal{S} = \{s_1, s_2, s_3\}$. Write out explicitly all elements of $\sigma(S_0)$ and $\sigma(S_0, S_1)$. Verify that $\sigma(S_0) \subseteq \sigma(S_0, S_1)$ (the filtration is increasing). For a value function $V : \mathcal{S} \to \mathbb{R}$, verify that $V(S_0)$ is $\sigma(S_0)$-measurable by identifying the function g in the Doob–Dynkin lemma.

8 Summary of Key Results

Result	**Statement**	**Role**
σ-algebra axioms	Closed under complement and countable union	Framework for events
Generated σ-algebra	$\sigma(\mathcal{C})$ = smallest σ-algebra $\supseteq \mathcal{C}$	Defines filtrations
π–λ theorem	π-system $\subseteq$ λ-system $\implies$ $\sigma(\pi) \subseteq \lambda$	Uniqueness of measures
Measure properties	Monotonicity, sub-additivity, continuity from above/below	All measure arguments
Carathéodory	Pre-measure on algebra extends to σ-algebra	Constructs Lebesgue measure
Generator criterion	Check measurability on generators only	Verifying random variables
Approximation theorem	Non-negative measurable = lim of simple $\uparrow$	Defines Lebesgue integral
Doob–Dynkin	Y is $\sigma(X)$-measurable iff $Y = g(X)$	Pull-out property

Next lesson: Integration theory — Lebesgue integral, expectation, monotone/dominated convergence theorems, L^p spaces. These build directly on the simple function approximation theorem to define $\mathbb{E}[X]$ rigorously.